THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 6: **Triangles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In $\triangle ABC$, $DE \parallel BC$, $AD = 1$ cm, $DB = 3$ cm, $AE = 2$ cm. Then EC (in cm) = 1

(a) 7 **(b)** 2 **(c)** 6 **(d)** 3

2. $\triangle ABC \sim \triangle DEF$ with areas 16 cm² and 36 cm². If $AB = 6$ cm, then DE (cm) = 1

(a) 9 **(b)** 27 **(c)** 6 **(d)** 12

3. The length of the diagonal of a square of side 3 cm, squared, is: 1

(a) 18 **(b)** 6 **(c)** 36 **(d)** 9

4. A man goes 15 m due east and then 20 m due north. His distance from the start (m) is: 1

(a) 25 **(b)** 35 **(c)** 625 **(d)** 5

5. In $\triangle ABC$, $AB = 9$, $BC = 12$, $AC = 15$. The triangle is: 1

(a) equilateral **(b)** not right-angled **(c)** isosceles right **(d)** right-angled

6. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 2 : 3$. Then $AE : AC$ is 2: 1

(a) 3 **(b)** 6 **(c)** 5 **(d)** 4

7. In $\triangle ABC$, $DE \parallel BC$, $AD = 3$ cm, $DB = 6$ cm, $AE = 6$ cm. Then EC (in cm) = 1

(a) 15 **(b)** 12 **(c)** 18 **(d)** 6

8. $\triangle ABC \sim \triangle DEF$ with areas 18 cm² and 32 cm². If $AB = 9$ cm, then DE (cm) = 1

(a) 9 **(b)** 12 **(c)** 15 **(d)** 48

9. The length of the diagonal of a square of side 5 cm, squared, is: 1

(a) 50 **(b)** 100 **(c)** 10 **(d)** 25

10. A man goes 24 m due east and then 32 m due north. His distance from the start (m) is: 1

(a) 1600 **(b)** 8 **(c)** 40 **(d)** 56

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A ladder 17 m long reaches a window 15 m high. Find how far the foot of the ladder is from the wall. 2

12. Sides of a triangle are 7 cm, 24 cm, 25 cm. State with reason whether it is right-angled. 2

13. The perimeters of two similar triangles are 20 cm and 40 cm. Find the ratio of their corresponding sides and their areas. 2

14. A ladder 25 m long reaches a window 24 m high. Find how far the foot of the ladder is from the wall. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 4$ cm, $DB = 6$ cm and $AE = 12$ cm, find EC. 3

16. The areas of two similar triangles are 16 cm² and 36 cm². If a side of the first is 4 cm, find the corresponding side of the second. 3

17. In $\triangle ABC$, $DE \parallel BC$ with $AD : DB = 1 : 2$. Find the ratio of the area of $\triangle ADE$ to that of $\triangle ABC$, and to that of trapezium $DBCE$. 3

OR

In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 5$ cm, $DB = 3$ cm and $AE = 10$ cm, find EC. 3

18. The areas of two similar triangles are 18 cm² and 162 cm². If a side of the first is 3 cm, find the corresponding side of the second. 3

OR

In $\triangle ABC$, $DE \parallel BC$ with $AD : DB = 2 : 3$. Find the ratio of the area of $\triangle ADE$ to that of $\triangle ABC$, and to that of trapezium $DBCE$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. 5

OR

Two similar triangles have perimeters 18 cm and 36 cm. If a side of the smaller is 5 cm, find the corresponding side of the larger and the ratio of their areas. 5

20. State and prove the converse of the Basic Proportionality Theorem. 5

OR

Two similar triangles have perimeters 20 cm and 60 cm. If a side of the smaller is 6 cm, find the corresponding side of the larger and the ratio of their areas. 5

END OF QUESTION PAPER Total: 40 Marks

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